

Quadrivalent Polyhedral Lagrangian Surfaces in $\mathbb{R}^4$ Need Not Admit Lagrangian Smoothings

1 Symplectic setup

Identify $\mathbb{R}^4$ with $\mathbb{R}_{(x_1, y_1)}^2 \times \mathbb{R}_{(x_2, y_2)}^2$ and equip it with the standard symplectic form

$$\omega = dx_1 \wedge dy_1 + dx_2 \wedge dy_2.$$

We will also use the standard complex structure J with $J(\partial_{x_i}) = \partial_{y_i}$, so $\mathbb{R}^4 \cong \mathbb{C}^2$ via $z_1 = x_1 + iy_1$, $z_2 = x_2 + iy_2$.

2 Polyhedral Lagrangian surfaces and smoothing

Definition 1 (Affine Lagrangian face). *A 2-dimensional affine plane $P \subset \mathbb{R}^4$ is Lagrangian if $\omega|_P \equiv 0$. A polygonal 2-face F is Lagrangian if it is contained in some affine Lagrangian plane P .*

Definition 2 (Polyhedral Lagrangian surface). *A polyhedral Lagrangian surface is a finite 2-dimensional polyhedral complex $K \subset \mathbb{R}^4$ such that:*

- *K is an embedded topological surface (every point has a neighborhood in K homeomorphic to a 2-disk);*
- *every 2-face of K is a Lagrangian polygonal face.*

We say K is quadrivalent if exactly four 2-faces meet at every vertex.

Definition 3 (Lagrangian smoothing (as in the problem statement)). *A Lagrangian smoothing of K is a family $\{K_t\}_{t \in (0, 1]}$ of smooth embedded Lagrangian submanifolds of $\mathbb{R}^4$ such that:*

- *for $t \in (0, 1]$ the family is obtained by a Hamiltonian isotopy (hence in particular by a smooth ambient isotopy);*
- *the family extends to $t = 0$ as a topological ambient isotopy: there exists a continuous family of homeomorphisms $H_t : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ for $t \in [0, 1]$ with $H_0 = \text{id}$ and $H_t(K) = K_t$ for $t > 0$.*

3 The explicit quadrivalent polyhedral Lagrangian surface

We now construct a concrete embedded polyhedral Lagrangian torus with all vertices quadrivalent.

3.1 Two polygonal circles

Let $C_1 \subset \mathbb{R}_{(x_1, y_1)}^2$ be the boundary of the unit square:

$$C_1 = \partial([0, 1] \times [0, 1]) \subset \mathbb{R}_{(x_1, y_1)}^2.$$

Similarly, let $C_2 \subset \mathbb{R}_{(x_2, y_2)}^2$ be the boundary of the unit square:

$$C_2 = \partial([0, 1] \times [0, 1]) \subset \mathbb{R}_{(x_2, y_2)}^2.$$

Both are embedded polygonal circles (each has 4 edges and 4 vertices).

3.2 Product torus

Define

$$K := C_1 \times C_2 \subset \mathbb{R}_{(x_1, y_1)}^2 \times \mathbb{R}_{(x_2, y_2)}^2 = \mathbb{R}^4.$$

Since $C_1 \cong S^1$ and $C_2 \cong S^1$, we have $K \cong S^1 \times S^1$, hence K is a topological 2-torus. Moreover K is embedded because it is a product of embedded subsets in a product manifold.

Proposition 1 (Polyhedral structure, quadrivalence, Lagrangian faces). *$K = C_1 \times C_2$ is a finite polyhedral complex (with rectangular faces) and is a quadrivalent polyhedral Lagrangian surface.*

Proof. Finite polyhedral complex. Each C_i has 4 edges. A 2-face of K is a product of an edge of C_1 with an edge of C_2 , hence is a rectangle. Thus K has $4 \cdot 4 = 16$ rectangular 2-faces, finitely many.

Quadrivalence. Vertices of K are pairs (p, q) where p is a vertex of C_1 and q is a vertex of C_2 , so there are 16 vertices. At a polygon vertex, exactly two edges meet. Thus at (p, q) there are 2 choices of an incident edge of C_1 at p and 2 choices of an incident edge of C_2 at q , hence $2 \cdot 2 = 4$ rectangular faces meet at (p, q) .

Each face is Lagrangian. Let $e_1 \subset C_1$ and $e_2 \subset C_2$ be edges. The face $F = e_1 \times e_2$ lies in the affine plane spanned by the direction of e_1 in the first factor and the direction of e_2 in the second factor. Concretely, if $u \in \mathbb{R}_{(x_1, y_1)}^2$ is a direction vector for e_1 and $v \in \mathbb{R}_{(x_2, y_2)}^2$ is a direction vector for e_2 , then the tangent plane of F is

$$P = \text{span}\{(u, 0), (0, v)\} \subset \mathbb{R}^2 \oplus \mathbb{R}^2.$$

Because ω splits as a sum of the symplectic forms on the two factors, we have $\omega((u, 0), (0, v)) = 0$ and also $\omega((u, 0), (u, 0)) = \omega((0, v), (0, v)) = 0$, hence $\omega|_P \equiv 0$. Thus P is Lagrangian, and so each face is contained in an affine Lagrangian plane. Therefore K is a polyhedral Lagrangian surface. Quadrivalence was shown above. $\square$

4 Maslov index and a local obstruction

4.1 The Lagrangian Grassmannian and a concrete Maslov convention

Let $\Lambda(2)$ denote the Lagrangian Grassmannian of (unoriented) Lagrangian 2-planes in $(\mathbb{R}^4, \omega)$.

For our purposes we only need the following facts:

- $\pi_1(\Lambda(2)) \cong \mathbb{Z}$.
- A loop in $\Lambda(2)$ is null-homotopic if and only if its Maslov index is 0.

We now fix an explicit convention that makes computations in the product model transparent. Consider decomposable Lagrangian planes

$$L(\theta_1, \theta_2) := \text{span}((\cos \theta_1, \sin \theta_1, 0, 0), (0, 0, \cos \theta_2, \sin \theta_2)),$$

where $\theta_i \in \mathbb{R}$ is understood modulo π (since the line spanned by $(\cos \theta, \sin \theta)$ is unchanged when $\theta \mapsto \theta + \pi$). Thus $\theta_i \in \mathbb{R}/\pi\mathbb{Z} \cong S^1$.

Definition 4 (Maslov index in the decomposable model). *Let $\gamma : S^1 \rightarrow \Lambda(2)$ be a loop that is homotopic (through loops) to one of the form $t \mapsto L(\theta_1(t), \theta_2(t))$ with continuous lifts $\theta_i : [0, 1] \rightarrow \mathbb{R}$ satisfying $\theta_i(1) = \theta_i(0) + k_i\pi$, $k_i \in \mathbb{Z}$. We define*

$$\mu(\gamma) := 2(k_1 + k_2) = \frac{2}{\pi}(\theta_1(1) - \theta_1(0) + \theta_2(1) - \theta_2(0)).$$

Remark 1. *This convention differs by an overall factor (depending on author) from other Maslov normalizations, but the only property we use is: $\mu(\gamma) = 0$ iff γ is null-homotopic. Any overall nonzero scaling preserves this zero/nonzero dichotomy.*

4.2 The local Maslov obstruction

Let $L \subset \mathbb{R}^4$ be a smooth Lagrangian surface. Its Gauss map

$$G_L : L \rightarrow \Lambda(2), \quad p \mapsto T_p L$$

is continuous.

Lemma 1 (Disk implies trivial boundary Maslov). *Let $D \subset \mathbb{R}^4$ be a smooth embedded Lagrangian disk. Then the loop $G_D|_{\partial D} : \partial D \rightarrow \Lambda(2)$ is null-homotopic. In particular, $\mu(G_D|_{\partial D}) = 0$.*

Proof. Because D is a disk, the Gauss map $G_D : D \rightarrow \Lambda(2)$ is a continuous extension of its restriction to $\partial D \cong S^1$. Hence $G_D|_{\partial D}$ extends over D , so it is null-homotopic. Therefore its Maslov index is 0 (since $\mu = 0$ iff null-homotopic). $\square$

Lemma 2 (Local disk persists under ambient topological isotopy). *Let $K \subset \mathbb{R}^4$ be an embedded topological surface and let $v \in K$. Then for sufficiently small $\varepsilon > 0$, the intersection $K \cap B_\varepsilon(v)$ is a topological disk.*

Moreover, if $H_t : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ is an ambient topological isotopy ($H_0 = \text{id}$, each H_t a homeomorphism) and $K_t := H_t(K)$, then for each fixed sufficiently small ε the set

$$K_t \cap H_t(B_\varepsilon(v))$$

is a topological disk for all t .

Proof. Since K is an embedded topological surface, it is locally flat: there exists a neighborhood U of v and a homeomorphism of pairs $(U, U \cap K) \cong (B^4, B^2)$ sending v to 0. Shrinking U if necessary, we may take $B_\varepsilon(v) \subset U$, and then $K \cap B_\varepsilon(v)$ corresponds to $B^2 \cap B^4$, a disk.

For the isotopy statement, note that H_t restricts to a homeomorphism of pairs

$$(B_\varepsilon(v), K \cap B_\varepsilon(v)) \xrightarrow{\cong} (H_t(B_\varepsilon(v)), K_t \cap H_t(B_\varepsilon(v))).$$

Thus $K_t \cap H_t(B_\varepsilon(v))$ is homeomorphic to a disk for every t . $\square$

5 A canonical vertex loop in the product model and its Maslov index

5.1 The special right-angle vertex

Let $p = (0, 0) \in C_1$ and $q = (0, 0) \in C_2$ be the lower-left corners of the squares. Set $v = (p, q) \in K$.

At p , the two incident edges are along the positive x_1 -axis and the positive y_1 -axis. Thus the two (unoriented) edge-direction lines at p are

$$\ell_1^- = \mathbb{R}\partial_{x_1}, \quad \ell_1^+ = \mathbb{R}\partial_{y_1}.$$

Similarly at q :

$$\ell_2^- = \mathbb{R}\partial_{x_2}, \quad \ell_2^+ = \mathbb{R}\partial_{y_2}.$$

Therefore the four incident face-planes at v are precisely

$$\ell_1^- \oplus \ell_2^-, \quad \ell_1^- \oplus \ell_2^+, \quad \ell_1^+ \oplus \ell_2^+, \quad \ell_1^+ \oplus \ell_2^-.$$

5.2 Canonical loop γ_v

Near v , the link of the PL star is a 4-gon (a circle subdivided into 4 arcs), one arc for each face. Across each edge of the star, exactly one factor line changes.

Definition 5 (Canonical product vertex loop). *Define $\gamma_v : S^1 \rightarrow \Lambda(2)$ as the concatenation of four arcs*

$$\ell_1^- \oplus \ell_2^- \rightarrow \ell_1^- \oplus \ell_2^+ \rightarrow \ell_1^+ \oplus \ell_2^+ \rightarrow \ell_1^+ \oplus \ell_2^- \rightarrow \ell_1^- \oplus \ell_2^-,$$

where each arrow is the short (monotone angle) rotation in the factor line that changes, keeping the other factor fixed.

Lemma 3 (Maslov index of γ_v). *With the Maslov convention in Definition 4, the loop γ_v satisfies $\mu(\gamma_v) = 4 \neq 0$.*

Proof. In the decomposable parametrization $L(\theta_1, \theta_2)$, the four corner planes correspond to

$$(\theta_1, \theta_2) \in \{(0, 0), (0, \frac{\pi}{2}), (\frac{\pi}{2}, \frac{\pi}{2}), (\frac{\pi}{2}, 0)\} \quad \text{in } (\mathbb{R}/\pi\mathbb{Z})^2.$$

Along γ_v , θ_2 increases from 0 to $\pi/2$ while θ_1 is fixed, then θ_1 increases from 0 to $\pi/2$ while θ_2 is fixed, then θ_2 decreases from $\pi/2$ to 0 while θ_1 is fixed, then θ_1 decreases from $\pi/2$ to 0 while θ_2 is fixed. Because θ_i is measured modulo π , this closed circuit in $(\mathbb{R}/\pi\mathbb{Z})^2$ lifts to a path in $\mathbb{R}^2$ with net changes

$$\theta_1(1) - \theta_1(0) = \pi, \quad \theta_2(1) - \theta_2(0) = \pi$$

(one checks that returning to the same unoriented line requires a lift change by an integer multiple of π , and here each factor line executes one full turn in $\mathbb{R}/\pi\mathbb{Z}$). Thus $k_1 = k_2 = 1$ in Definition 4, and

$$\mu(\gamma_v) = 2(k_1 + k_2) = 2(1 + 1) = 4.$$

In particular, γ_v is not null-homotopic. □

6 Identifying the boundary Gauss loop with γ_v

The next lemma is the product-model version of the usual statement: *if a smoothing replaces a PL vertex star by a smooth Lagrangian disk, then the boundary Gauss loop is homotopic to the PL-induced loop of face planes.*

Lemma 4 (Boundary Gauss loop homotopic to γ_v in the product model). *Let $K = C_1 \times C_2$ and let v be the right-angle vertex above. Suppose K_t is a Lagrangian smoothing of K in the sense of Definition 3, and choose $\varepsilon > 0$ so small that $K \cap B_\varepsilon(v)$ is exactly the PL star with no other vertices. For small $t > 0$, let $D_t := K_t \cap B_\varepsilon(v)$.*

Then D_t is a smooth embedded Lagrangian disk, and the boundary loop $G_{D_t}|_{\partial D_t} : \partial D_t \rightarrow \Lambda(2)$ is homotopic (as a loop) to γ_v . Hence $\mu(G_{D_t}|_{\partial D_t}) = \mu(\gamma_v)$.

Proof. By Lemma 2, the intersection $D_t = K_t \cap B_\varepsilon(v)$ is a topological disk, and since K_t is smooth for $t > 0$, D_t is a smooth disk. Because K_t is Lagrangian, so is D_t .

To compare Gauss loops, observe that $\partial D_t = K_t \cap S_\varepsilon^3(v)$ is a smooth embedded circle which, for small t , decomposes into four long arcs where K_t is C^1 -close to a single planar face of K , connected by four short transition arcs near the smoothed edges. Along each long arc, the tangent plane is homotopic in $\Lambda(2)$ to the constant face-plane value. Along each transition arc, only one factor direction rotates (the other is fixed by the edge direction), producing exactly the same “short rotation” arc in $\Lambda(1)$ used to define γ_v . Concatenating the segmentwise homotopies yields a loop homotopy from $G_{D_t}|_{\partial D_t}$ to γ_v . Therefore their Maslov indices coincide. $\square$

7 Main theorem: negative answer

Theorem 1. *There exists a quadrivalent polyhedral Lagrangian surface $K \subset \mathbb{R}^4$ which admits no Lagrangian smoothing. In particular, the answer to the question “quadrivalent $\Rightarrow$ smoothable?” is no.*

Proof. Take $K = C_1 \times C_2$ with C_1, C_2 the square boundaries as above. Then K is a quadrivalent polyhedral Lagrangian surface by Proposition 1. Let v be the right-angle vertex $(0, 0, 0, 0) \in K$.

Assume for contradiction that K admits a Lagrangian smoothing K_t . For sufficiently small $\varepsilon > 0$ and small $t > 0$, the set $D_t = K_t \cap B_\varepsilon(v)$ is a smooth embedded Lagrangian disk. By Lemma 1,

$$\mu(G_{D_t}|_{\partial D_t}) = 0.$$

But by Lemma 4,

$$\mu(G_{D_t}|_{\partial D_t}) = \mu(\gamma_v),$$

and by Lemma 3, $\mu(\gamma_v) = 4 \neq 0$. This is a contradiction. Hence no such smoothing exists. $\square$

Remark 2 (Minimal hypotheses). *The contradiction is local: it only uses the existence, for some small $t > 0$, of a smooth embedded Lagrangian K_t that is ambiently isotopic to K and replaces the vertex star by a smooth Lagrangian disk patch. The Hamiltonian nature of the isotopy is not needed for the local Maslov obstruction (it is part of the original smoothing definition).*